E&ICT, IIT Roorkee



Advanced Program in

Applied Data Analytics & Generative AI

By E&ICT Academy, IIT Roorkee

6-Months 140 Hours (Curriculum)

About The IIT Roorkee

Indian Institute of Technology Roorkee (IIT Roorkee), one of the oldest technical institutions in Asia, was established in 1847 as the Thomason College of Civil Engineering and was later converted to an IIT in 2001.

The institute offers a wide range of academic programmes including BTech, BArch, BS, MTech, MSc, MBA, MDes, and PhD across its 23 academic departments, centres, and schools covering engineering, sciences, architecture, management, humanities, and social sciences.

Over the years, IIT Roorkee has developed state-of-the-art infrastructure, a strong research ecosystem, and a rich legacy of excellence in technical education and innovation. The institute continues to be recognized nationally and internationally for its academic rigour, research output, and contributions to industry and Society.



About The E&ICT Academy IIT Roorkee

Electronics & ICT Academy, IIT Roorkee, aims to strengthen the skills of faculty members, working professionals, and students from Engineering, Science, Commerce, and Polytechnic institutions through specialized short-term training programs.

The Academy focuses on both fundamental and advanced areas of Information Technology, Electronics & Communication, Product Design, and Manufacturing.

In addition, E&ICT Academy, IIT Roorkee, organizes customized industry-oriented training programs, faculty development programs (FDPs), and research promotion workshops to bridge the gap between academia and industry. These initiatives are designed to enhance technical competency, encourage innovation, and promote collaborative research in emerging technologies.



About The IoT Academy

The IoT Academy, established in 2017 is a fast emerging company imparting quality programs for skills training, internship and guidance in cutting edge technologies like Data Science, Machine Learning, Artificial Intelligence, Internet of Things, Embedded Systems & many more and focused on helping people develop the skills they need to thrive in the rapidly growing digital economy.

The IoT Academy has collaborated with various premier institutes e.g. E&ICT Academy, IIT-Roorkee, IIT-Guwahati and IIT-Kanpur for Advance Certification courses to take provide outcome-centric solutions to help them achieve their professional goals.

400+
Hiring Partners

30,000+
Learners

8+ Years
Of excellence

81%
Salary Hike

91%
Happy Outcomes



About The Program

Dive into the exciting world of Applied Data Analytics and Generative AI. This program helps you learn how to study data, understand it, and use it to make better decisions. You will also learn how generative AI works and how it can create new ideas and solutions. These skills can be used in many fields like healthcare, finance, retail, and technology. This course gives you the knowledge to use data and AI to solve real problems in the real world.



Mastering Data Analytics & Generative AI

- Learn the basics of data analytics. You will understand how to collect data, clean it, and show it in charts. You will also learn simple statistics.
- Generative AI helps by making new data, giving better predictions, and doing some tasks automatically.

Practical Applications

- You will work on real projects. You will explore data, build prediction models, check risks, and test different situations.
- Generative AI will help you find useful insights, work faster, and solve business problems.

Industry Trends & Real-World Impact

- You will study real examples from companies like Google, Microsoft, and IBM.
- You will learn how data analytics and generative AI are changing industries. They help companies make better decisions, give personalized services, and create new ideas.

Ethics, Governance, and Responsible AI

- You will learn about common issues like bias, data privacy, and copyright in AI.
- You will understand how to use AI in a safe and fair way by following rules and protecting data.

Introduction to Data Analytics & Generative AI (3 Hours 1 Session)

- Overview of Data Analytics
- Overview of Artificial Intelligence
- Traditional AI vs. Generative AI
- Evolution of AI models (Rule-Based → ML → DL → Generative AI)
- Industry Use Cases

Data Analytics & Data Visualization with Python and Business Analytics (45 hours 15 Sessions)

1. **Python Basics**
   - What is Python
   - Application of Python
   - Why use Python for AI-ML Installation
   - Anaconda/Other Idle
   - Python Tokens
   - Data Types in Python
   - Conditional Statement
   - Loops in Python
   - Functions in Python
   - Advance Functions
   - File Handling
   - Mini Project

2. **Intro to Git & Github and Kaggle**
   - Git and GitHub
   - Kaggle
3. **Python NumPy & Pandas**
   - Introduction to NumPy
   - Exploring a NumPy Array
   - Indexing & Slicing a NumPy Array
   - Manipulating a NumPy Array
   - Performing Mathematical & Statistical Functions using NumPy
   - Performing Linear Algebra Operations using NumPy
   - Introduction to Pandas
   - Exploring Pandas Series
   - Introduction to Pandas DataFrame
   - Importing & Exporting Data
   - Implementing basic DataFrame functionalities
   - Exploring Descriptive Statistics with Pandas
4. **Exploratory Data Analysis (EDA)**
   - Data Types
   - Dispersion & Skewness
   - Data imputation
   - Data Pre-processing
   - Data Cleaning
   - Data Manipulation
   - Advanced Manipulation
5. **SQL For Data Analytics**
   - SQL Basic
   - SQL Joins
   - SQL Aggregations
   - Subqueries and Temp Tables
   - SQL Data Cleaning
   - Window Functions

6. **Data Analysis with Excel**
   - Key Formulas and Functions, Ranges and Tables
   - Data Cleaning – Text Functions, Dates, and Times
   - Conditional Formatting, Sorting, and Filtering
   - Dashboard Creation
   - Analysis with Pivot Tables
   - Data Analysis in Excel – Trends and Patterns
   - Data Visualization in Excel – Charts and Plots
   - Functions in Python
   - Advance Functions
   - File Handling
7. **Business Analytics**
   - Business Analysis in a Data + AI World
   - Requirement Gathering for Analytics & AI
   - Documentation Essentials
   - Tools for Business Analytics
   - Process Analysis & KPI Design
   - Data Interpretation & Insight Storytelling
   - Designing GenAI Solutions
   - Agile Business Analysis Essentials
   - Testing & Validation (UAT)
   - Real-World Case Studies Domain Wise
8. **Data Visualization with Python**
   - Why Data Visualization?
   - Introduction to Data Visualization
   - Libraries & Tools for Data Visualization in Python
   - Static Data Visualization Using Seaborn
   - Interactive Data Visualization Using Plotly Express
   - Interactive Animations & Facet Plots

9. **Tableau for Business Intelligence**
    - Understanding Data
    - Creating Your First Visualization
    - Tableau Calculations
    - Formatting Visualizations
    - Manipulating Data in Tableau
    - Creating Dashboards & Stories
    - Distributing & Publishing Your Visualization
10. **Visual Storytelling using Power BI**
    - Understanding Data
    - Creating Your First Visualization
    - Tableau Calculations
    - Formatting Visualizations
    - Manipulating Data in Tableau
    - Creating Dashboards & Stories
    - Distributing & Publishing Your Visualization

Mathematics Fundamentals (15 hours 4 Sessions)

- Fundamentals of Algebra
    - Vectors, Matrices, Eigenvalues & Eigenvectors
    - Dot Product, Matrix Operations, Applications in AI
- Probability & Statistics
    - Descriptive Statistics (Mean, Median, Mode, Variance, Standard Deviation)
    - Probability Distributions: Normal, Bernoulli, Binomial
    - Central Limit Theorem
- Fundamentals of Calculus

NLP Foundation - 15 hours (5 Sessions)

- Artificial Neural Network (ANN)
- Convolutional Neural Network (CNN)
- Recurrent Neural Network (RNN)

Foundations of Generative AI & Prompt Engineering (12 Hours 4 Sessions)

Introduction to Generative AI

- What is Generative AI? Evolution from Rule-Based AI to GenAI
- Differences: Generative AI vs. Traditional AI
- Industry Applications:
  - Content Creation (Chatbots, Story Writing, Blogging)
  - Code Generation (AI-Assisted Programming)
  - Image & Video Synthesis (DALL·E, MidJourney, DeepSeek Vision)

Prompt Engineering & Optimization

- Fundamentals of Prompt Engineering
  - Zero-shot, Few-shot, Chain-of-Thought (CoT) Prompting
  - Role-based Prompts, Multi-turn Conversations
- Advanced Prompting Techniques
  - System vs. User Prompts, Style Transfer, and Iterative Prompting

Generative AI Architectures & Large Language Models (LLMs) - 24 Hours (8 Sessions)

Encoder-Decoder Architectures & Attention Mechanism

- Understanding Sequence-to-Sequence (Seq2Seq) Models
- Introduction to Attention Mechanisms
  - Self-Attention & Cross-Attention
  - Scaled Dot-Product Attention & Multi-Head Attention
- Transformers & Evolution
  - Introduction to Transformers
  - Positional Encoding & Feed-Forward Networks

Foundation of Generative Models

- Types of Generative Models
  - Variational Autoencoders (VAEs)
  - Generative Adversarial Networks (GANs) – StyleGAN, CycleGAN
  - Transformer-Based Models (GPT, BERT, T5)
- Understanding AI Hallucinations, Bias, and Model Safety
  - Variational Autoencoders (VAEs)
  - Generative Adversarial Networks (GANs) – StyleGAN, CycleGAN
  - Transformer-Based Models (GPT, BERT, T5)

Large Language Models (LLMs) & Fine-Tuning

- Popular LLMs: GPT-4, DeepSeek, LLaMA, Mistral, Claude, Falcon
- Fine-Tuning vs. Pre-training of LLMs
  - Transfer Learning for LLMs
  - Customizing LLMs for Domain-Specific Use Cases

Retrieval-Augmented Generation (RAG) & Multimodal AI - 18 hours (6 Sessions)

Retrieval-Augmented Generation (RAG)

- Why RAG? Understanding LLM Limitations
- RAG Pipeline Components:
  - Document Embeddings & Vector Databases (Pinecone, ChromaDB)
  - Query Processing & Context Retrieval

Foundation of Generative Models

- Implementing RAG with Hugging Face & LangChain
- What is LlamaIndex?
- Using LlamaIndex for Efficient Document Retrieval

Multimodal AI – Text, Image, Video & Speech Processing

- Text-to-Image & Image-to-Image Generation
  - How DALL·E, MidJourney, and Stable Diffusion Work
  - Latent Diffusion Models (LDMs) & Hugging Face Implementations
- Speech-to-Text & AI Voice Processing
  - Whisper, Wav2Vec, DeepSeek-STT, Real-Time Speech Transcription
- Advanced Multimodal Models
  - OpenAI GPT-4V, Gemini
  - Text-to-Image, Image Captioning

Agentic AI & Hugging Face Ecosystem - 15 Hours (5 Sessions)

Agentic AI & Autonomous AI Agents

- What is Agentic AI?
  - Understanding Autonomous AI Systems
- AI Agent Frameworks:
  - LangChain, AutoGPT, Hugging Face Agents
  - Building AI Agents for Research, Trading, and Business Automation

Hugging Face Ecosystem

- Hugging Face Models, Spaces, and Transformers
- Fine-Tuning LLMs & Custom AI Deployments
- Enterprise Use Cases
  - AI Chatbots, Code Generation, AI for Research

Real-Time Generative AI in Business Applications

- Generative AI in Healthcare
  - AI-Powered Drug Discovery, Medical Imaging
- AI in Finance
  - Synthetic Data for Risk Modeling & Fraud Detection
- Automating Business Processes with AI

Module 1 Module 2 Module 3 Module 4 Module 5 Module 6 Module 7
Module 8 **Module 9** Module 10

Future of AI & Capstone Project - 5 Hours (2 Sessions)

Ethics, AI Governance & Future Trends

- Responsible AI & AI Governance
  - EU AI Act, OpenAI's AI Policy
- The Future of Generative AI
  - AI-Powered Autonomous Systems
  - Trends in AGI (Artificial General Intelligence)

Module 1 Module 2 Module 3 Module 4 Module 5 Module 6 Module 7
Module 8 Module 9 **Module 10**

Select A Domain That Aligns With Your Goals - 3 Hours (1 Sessions)

- Generative AI in Marketing
- Generative AI in Healthcare
- Generative AI in Legal Services
- Generative AI in Retail and E-Commerce
- Generative AI in Finance
- Generative AI in Education
- Generative AI in Human Resources (HR)
- Generative AI for Leadership Roles

Machine Learning Elective (Optional | Self-Paced)

- Introduction to Machine Learning
- Regression
- Supervised Classification
- Unsupervised Learning
- Dimension Reduction
- Association Rules Mining
- Recommendation System
- Time-series Forecasting
- Machine Learning Model Deployment

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